4 Convolutions

Time invariance:

$$y(t) = \sin(t)x(t)$$

is not time invariant. If $x(t) \to x(t-t_0)$, then we would have

$$y(t) = \sin(t)x(t-t_0)$$

however,

$$y(t-t_0) = \sin(t-t_0)x(t-t_0)$$

which is not the same.

We define a convolution

$$x[n] \circledast h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = y[n]$$

In continuous time

$$x \circledast h = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$